

# 1. Input

Section, materials and resistance basis exactly as analysed.

Project information

Project Name

Client

Company

Designed by

Address

PM-advanced (7) 2D reference case

Reference verification

P-M Column Designer

Engineering example

Seoul, Korea

Checked by

Date

Independent review

2026-07-23

Calculation identity

Section

Mechanics

Profile edition

Concrete model basis

Units

PM-advanced (7) 2D — 1500x1200 .

Stress-strain integration

2024 partial-revision framework (effe.

Code-default

mm · N · MPa; forces reported in kN,.

Calculation profile

Resistance profile

Verification status

Method id

Sign convention

KDS 2024 — Stress-strain integratio.

KDS 2024 current set · resistance: K.

draft

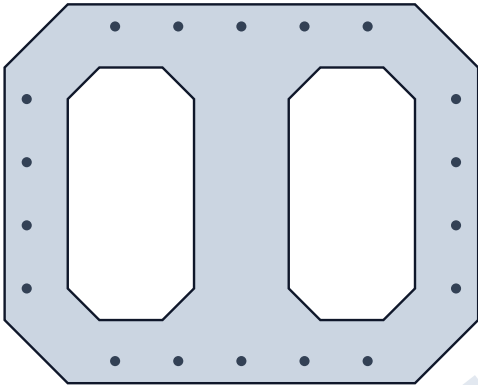
kds-2024-current-set-global-strength.

Compression positive; Mx = ΣF·(y-yc.

Section

Cross-section (mm, about the net concrete centroid)

18 bars



Section properties

Gross concrete area Ag

Reinforcement area As

Reinforcement ratio ρ

Bar count

Net centroid

Bounding box

Solids / holes

Perimeter

1,120,000 mm²

14,476 mm²

1.293 %

18

(0.0, 0.0) mm

1,500.0 × 1,200.0 mm

1 / 2

9,262.7 mm

Materials

Name

fck

εcu

α

Tension

Standard

Es

εy

Grades used

KDS C30 (workbook Input)

30.0 MPa

0.00330

0.8500

ignored

KDS

200,000 MPa

0.002000

1

Standard

Model

εc0

Ec

Name

fy

Model

εu

KDS

kds-parabolic

0.00200

28,417 MPa

SD400 (workbook Input)

400.0 MPa

elastic-perfectly-plastic

not declared

Design resistance basis

Format

φ compression (ties)

φ tension

Transition rule

Global resultant strength-reduction f.

0.650

0.850

fixed-or-yield-multiple

Transverse reinforcement

φ compression (spiral)

εt,limit

Axial cap

other

0.700

0.00500

0.800 × factored compression pole

Sampling and numerical evidence

Sampling mode

Active Design directions

Active Nominal points

Active Nominal stations

Worst interpolation error

Concrete discretization

fixed

36

972

27

— %

864 cells / 5196 points

Active Design points

Active Design stations

Active Nominal directions

Direction tolerance

Tolerance reached

Strain domain

972

27

36

— %

yes

concrete-pivot-ultimate

## Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is integrated from the material stress-strain law over a verified triangle/quadrature mesh of the section.

Overview interaction diagrams cut the active fixed 27-station, 10° direction surface. An arbitrary exact direction is calculated only when it is requested.

**Warning: Design profile status is draft; results are for review, not release.**

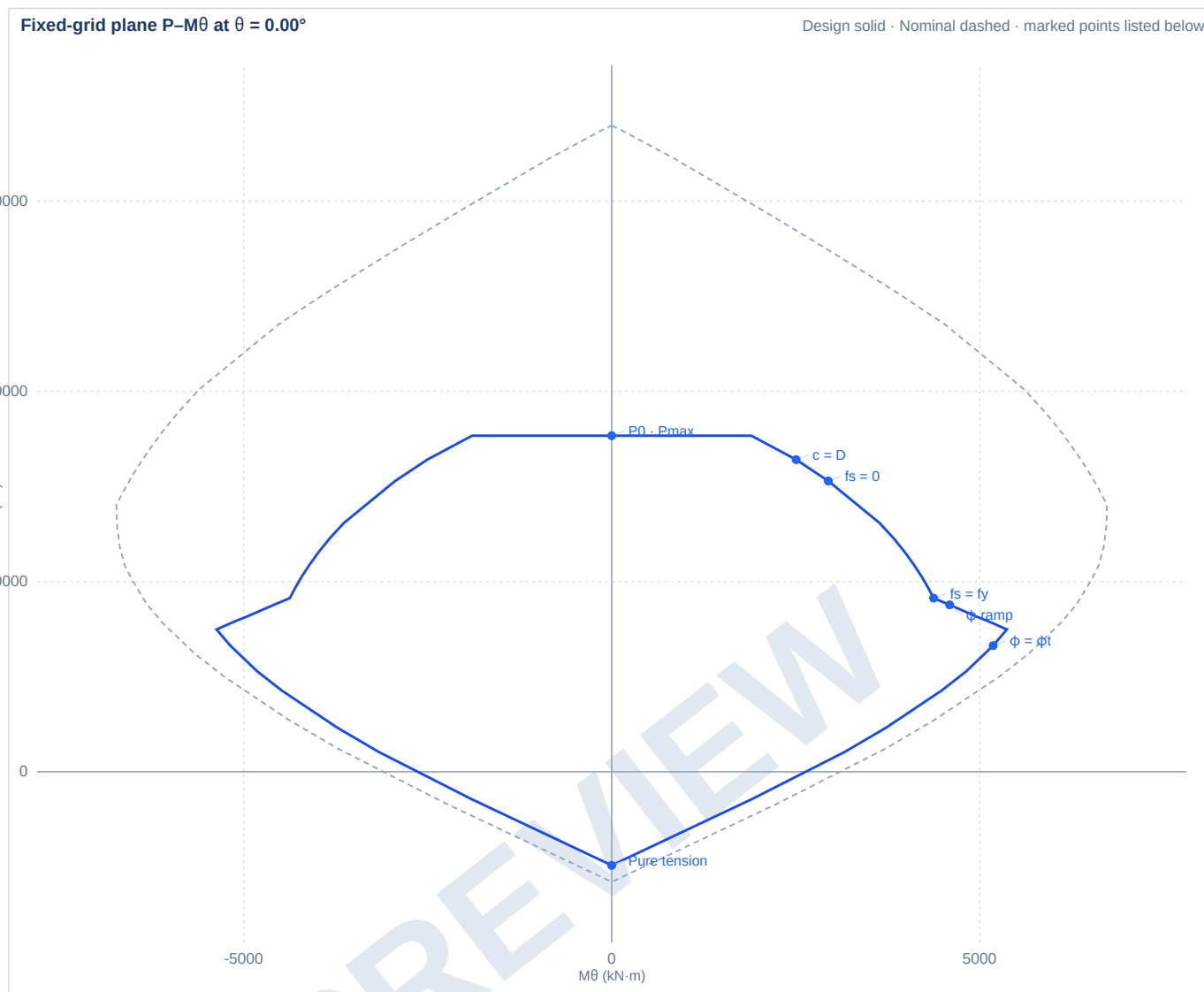
## Reinforcement schedule

| #  | Bar id | Ø (mm) | X (mm) | Y (mm) | As (mm²) | Steel                  |
|----|--------|--------|--------|--------|----------|------------------------|
| 1  | 1      | 32.0   | -400.0 | 530.0  | 804.2    | SD400 (workbook Input) |
| 2  | 2      | 32.0   | -200.0 | 530.0  | 804.2    | SD400 (workbook Input) |
| 3  | 3      | 32.0   | 0.0    | 530.0  | 804.2    | SD400 (workbook Input) |
| 4  | 4      | 32.0   | 200.0  | 530.0  | 804.2    | SD400 (workbook Input) |
| 5  | 5      | 32.0   | 400.0  | 530.0  | 804.2    | SD400 (workbook Input) |
| 6  | 6      | 32.0   | -400.0 | -530.0 | 804.2    | SD400 (workbook Input) |
| 7  | 7      | 32.0   | -200.0 | -530.0 | 804.2    | SD400 (workbook Input) |
| 8  | 8      | 32.0   | 0.0    | -530.0 | 804.2    | SD400 (workbook Input) |
| 9  | 9      | 32.0   | 200.0  | -530.0 | 804.2    | SD400 (workbook Input) |
| 10 | 10     | 32.0   | 400.0  | -530.0 | 804.2    | SD400 (workbook Input) |
| 11 | 11     | 32.0   | -680.0 | 300.0  | 804.2    | SD400 (workbook Input) |
| 12 | 12     | 32.0   | -680.0 | 100.0  | 804.2    | SD400 (workbook Input) |
| 13 | 13     | 32.0   | -680.0 | -100.0 | 804.2    | SD400 (workbook Input) |
| 14 | 14     | 32.0   | -680.0 | -300.0 | 804.2    | SD400 (workbook Input) |
| 15 | 15     | 32.0   | 680.0  | 300.0  | 804.2    | SD400 (workbook Input) |
| 16 | 16     | 32.0   | 680.0  | 100.0  | 804.2    | SD400 (workbook Input) |
| 17 | 17     | 32.0   | 680.0  | -100.0 | 804.2    | SD400 (workbook Input) |
| 18 | 18     | 32.0   | 680.0  | -300.0 | 804.2    | SD400 (workbook Input) |

## 2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.

Maximum design axial resistance applied at 17670.3 kN; it appears as the flat top of the diagram.



### Marked states on the diagram

| Mark            | Station criterion              | P design (kN) | M $\theta$ design (kN·m) | P nominal (kN) | M $\theta$ nominal (kN·m) | $\phi$ | Class                  |
|-----------------|--------------------------------|---------------|--------------------------|----------------|---------------------------|--------|------------------------|
| P0 · Pmax       | Pure compression               | 17670.3       | 0.00                     | 33981.4        | -0.00                     | 0.650  | compression-controlled |
| c = D           | c/D = 1                        | 16411.5       | 2505.81                  | 25248.5        | 3855.09                   | 0.650  | compression-controlled |
| fs = 0          | $\epsilon_s/\epsilon_y = 0$    | 15287.4       | 2943.07                  | 23519.0        | 4527.80                   | 0.650  | compression-controlled |
| fs = fy         | $\epsilon_s/\epsilon_y = 1$    | 9127.6        | 4374.42                  | 14042.5        | 6729.88                   | 0.650  | compression-controlled |
| $\phi$ ramp     | $\epsilon_s/\epsilon_y = 1.25$ | 8775.6        | 4591.67                  | 12842.3        | 6719.52                   | 0.683  | transition             |
| $\phi = \phi_t$ | $\epsilon_s/\epsilon_y = 3$    | 6636.6        | 5183.55                  | 7807.8         | 6098.30                   | 0.850  | tension-controlled     |
| Pure tension    | Pure tension                   | -4922.0       | 0.00                     | -5790.6        | 0.00                      | 0.850  | tension-controlled     |

P0 is the compression pole, Pmax the axial cap where one applies, fs = 0 the state where the extreme bar first carries no stress, and fs = fy the balanced state where it first yields. A  $\phi$  mark is the first station of a new resistance class, named in full in the Class column. Every other station on this meridian is tabulated per combination in section 5.

Other published directions (0.00°, 20.56°, 21.08°, 90.00°) are cut from the same surface; each combination is checked in section 4 against the plane it actually lies in.

3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

| # | Combination                | Pu (kN) | Mux (kN·m) | Muy (kN·m) | θL (°) | φ     | Class           | UR    | Verdict   |
|---|----------------------------|---------|------------|------------|--------|-------|-----------------|-------|-----------|
| 1 | ULS-1 workbook Newton d.   | 24942.9 | 3714.2     | 1431.8     | 21.1   | 0.650 | compression-co. | 1.539 | INADEQUA. |
| 2 | ULS-2 same axial, half mo. | 24942.9 | 1857.1     | 715.9      | 21.1   | 0.650 | compression-co. | 1.412 | INADEQUA. |
| 3 | ULS-3 low axial, high mom. | 5000.0  | 4000.0     | 1500.0     | 20.6   | 0.850 | transition      | 0.877 | ADEQUATE  |

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

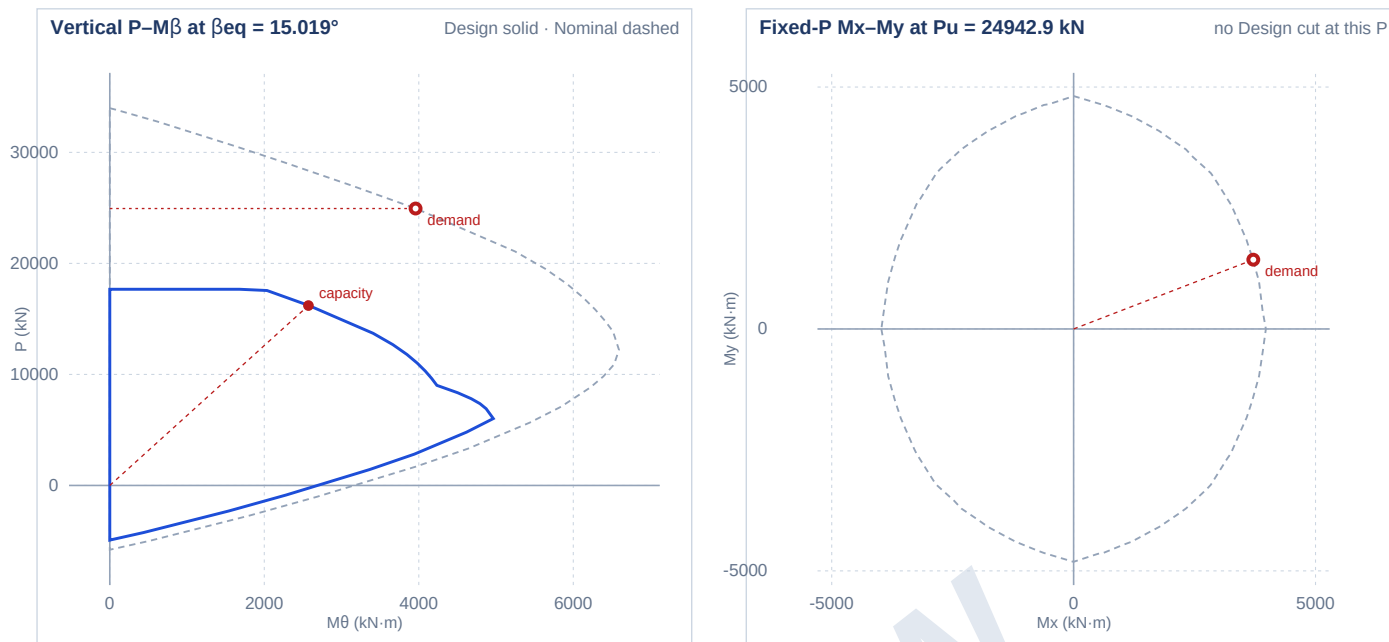
## 4.1 ULS-1 workbook Newton demand (on the 15 deg surface)

$P_u = 24942.9 \text{ kN}$  ·  $M_{ux} = 3714.2 \text{ kN}\cdot\text{m}$  ·  $M_{uy} = 1431.8 \text{ kN}\cdot\text{m}$  ·  $\theta_L = 21.08^\circ$

**INADEQUATE · utilization 1.5394 · fixed-P diagnostic —**

Converged preview equilibrium inside the material domain.

### Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

### Check parameters and solved state

|                               |                        |                             |              |
|-------------------------------|------------------------|-----------------------------|--------------|
| $P_u$                         | 24,942.9 kN            | $M_{ux}$                    | 3,714.2 kN·m |
| $M_{uy}$                      | 1,431.8 kN·m           | Moment direction $\theta_L$ | 21.08°       |
| Resultant moment              | 3,980.6 kN·m           | Strain-plane $\epsilon_0$   | 1.5162e-3    |
| Curvature $\kappa_x$          | 2.3759e-6              | Curvature $\kappa_y$        | 6.3748e-7    |
| Strain direction $\beta_{eq}$ | 15.019°                | Neutral-axis depth $c$      | 1,338.4 mm   |
| Capacity at $P_u$             | —                      | $\phi$                      | 0.6500       |
| Classification                | compression-controlled | Utilization UR              | 1.5394       |
| Fixed-P ratio                 | —                      | Verdict                     | INADEQUATE   |

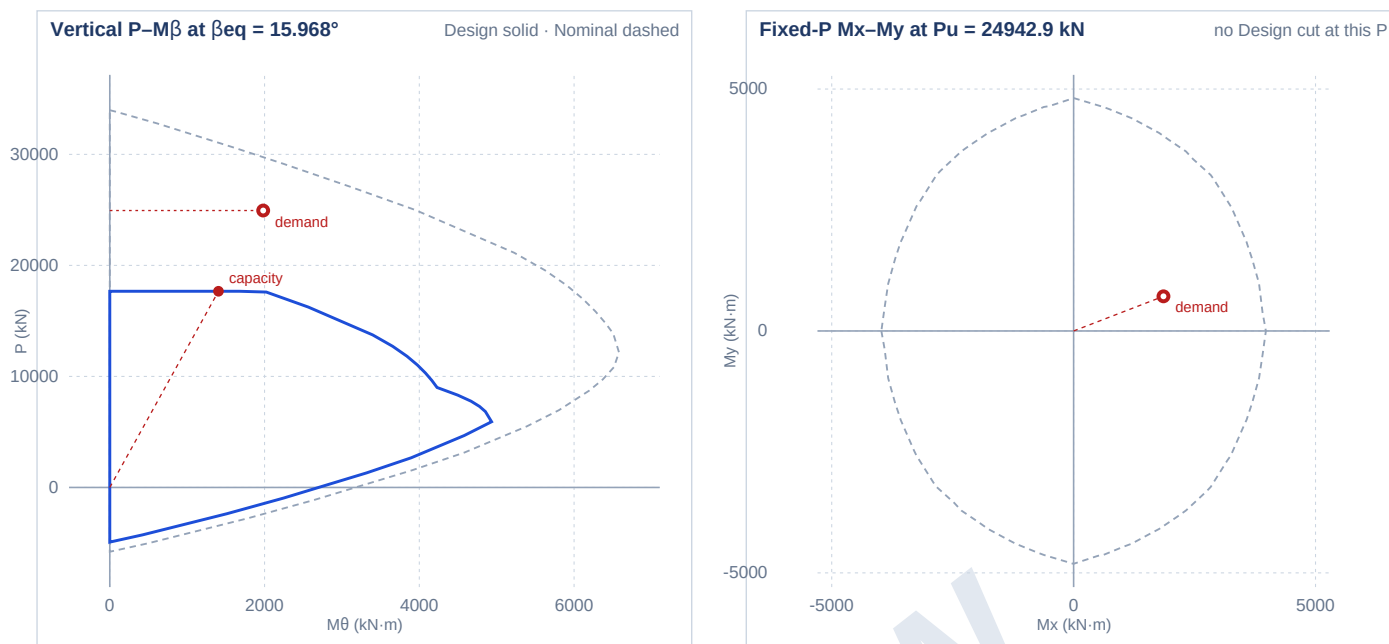
## 4.2 ULS-2 same axial, half moment (inside)

$P_u = 24942.9 \text{ kN}$  ·  $M_{ux} = 1857.1 \text{ kN}\cdot\text{m}$  ·  $M_{uy} = 715.9 \text{ kN}\cdot\text{m}$  ·  $\theta_L = 21.08^\circ$

**INADEQUATE · utilization 1.4116 · fixed-P diagnostic —**

Converged preview equilibrium inside the material domain.

### Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

### Check parameters and solved state

|                               |                        |                             |              |
|-------------------------------|------------------------|-----------------------------|--------------|
| $P_u$                         | 24,942.9 kN            | $M_{ux}$                    | 1,857.1 kN·m |
| $M_{uy}$                      | 715.9 kN·m             | Moment direction $\theta_L$ | 21.08°       |
| Resultant moment              | 1,990.3 kN·m           | Strain-plane $\epsilon_0$   | 1.0984e-3    |
| Curvature $\kappa_x$          | 7.8173e-7              | Curvature $\kappa_y$        | 2.2369e-7    |
| Strain direction $\beta_{eq}$ | 15.968°                | Neutral-axis depth $c$      | 2,079.0 mm   |
| Capacity at $P_u$             | —                      | $\phi$                      | 0.6500       |
| Classification                | compression-controlled | Utilization UR              | 1.4116       |
| Fixed-P ratio                 | —                      | Verdict                     | INADEQUATE   |

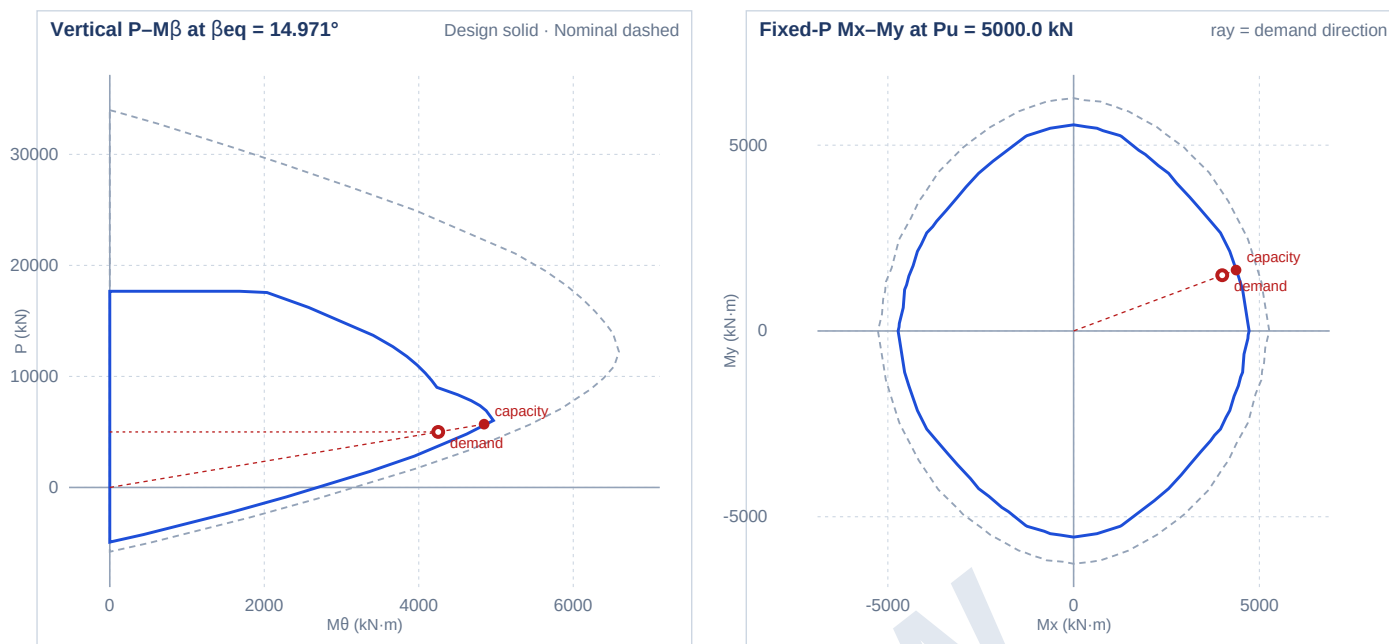
### 4.3 ULS-3 low axial, high moment

$P_u = 5000.0 \text{ kN}$  ·  $M_{ux} = 4000.0 \text{ kN}\cdot\text{m}$  ·  $M_{uy} = 1500.0 \text{ kN}\cdot\text{m}$  ·  $\theta_L = 20.56^\circ$

**ADEQUATE** · utilization **0.8774** · fixed-P diagnostic **0.9138**

Converged preview equilibrium inside the material domain.

#### Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

#### Check parameters and solved state

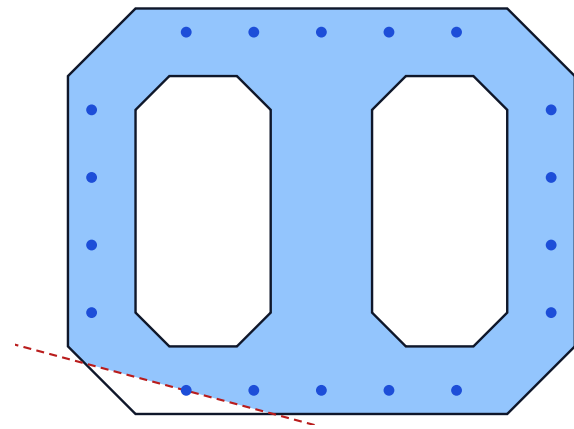
|                               |                                                                           |                             |                     |
|-------------------------------|---------------------------------------------------------------------------|-----------------------------|---------------------|
| $P_u$                         | <b>5,000.0 kN</b>                                                         | $M_{ux}$                    | <b>4,000.0 kN·m</b> |
| $M_{uy}$                      | <b>1,500.0 kN·m</b>                                                       | Moment direction $\theta_L$ | <b>20.56°</b>       |
| Resultant moment              | <b>4,272.0 kN·m</b>                                                       | Strain-plane $\epsilon_0$   | <b>-3.8060e-4</b>   |
| Curvature $\kappa_x$          | <b>2.5504e-6</b>                                                          | Curvature $\kappa_y$        | <b>6.8197e-7</b>    |
| Strain direction $\beta_{eq}$ | <b>14.971°</b>                                                            | Neutral-axis depth $c$      | <b>577.5 mm</b>     |
| Capacity at $P_u$             | <b><math>M_x 4,372.1 \cdot M_y 1,639.5 \text{ kN}\cdot\text{m}</math></b> | $\phi$                      | <b>0.8500</b>       |
| Classification                | <b>transition</b>                                                         | Utilization UR              | <b>0.8774</b>       |
| Fixed-P ratio                 | <b>0.9138</b>                                                             | Verdict                     | <b>ADEQUATE</b>     |

## 5.1 ULS-1 workbook Newton demand (on the 15 deg surface)

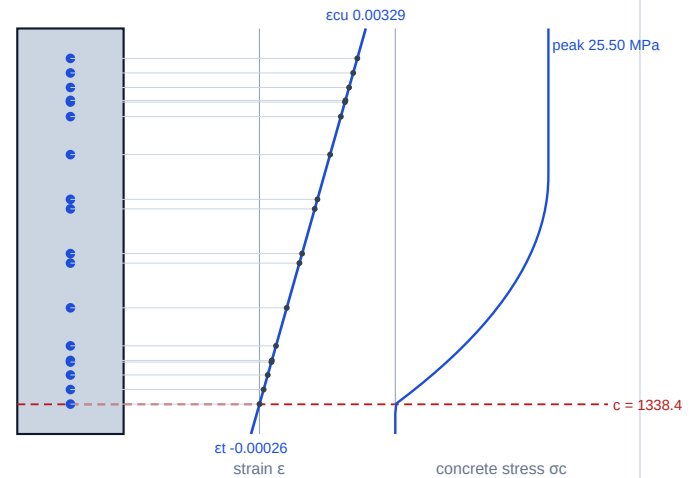
Worked section state and the points behind both curves.  $P_u = 24942.9 \text{ kN}$  ·  $M_{ux} = 3714.2 \text{ kN}\cdot\text{m}$  ·  $M_{uy} = 1431.8 \text{ kN}\cdot\text{m}$  ·  $\theta_L = 21.08^\circ$

### Section state

Transverse section — neutral axis and compression zone  $\varepsilon > 0$



Longitudinal section — strain and stress over the depth  $D\theta=1444$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

### Concrete and resultants

|                         |              |                         |              |
|-------------------------|--------------|-------------------------|--------------|
| Concrete force          | 21,577.4 kN  | Concrete Mx             | 2,900.4 kN·m |
| Concrete My             | 1,122.3 kN·m | Steel force (gross)     | 3,629.5 kN   |
| Displaced concrete      | -264.0 kN    | Steel net               | 3,365.5 kN   |
| Response P              | 24,942.9 kN  | Response Mx             | 3,714.2 kN·m |
| Response My             | 1,431.8 kN·m | Residual $ \Delta P $   | 1.01e-4 N    |
| Residual $ \Delta M_x $ | 3.15e-3 N·mm | Residual $ \Delta M_y $ | 4.99e-3 N·mm |

### Reinforcement ledger at the solved state

| Bar | X (mm) | Y (mm) | As (mm <sup>2</sup> ) | depth (mm) | $\varepsilon$ | $\sigma_s$ (MPa) | F (kN) | in block |
|-----|--------|--------|-----------------------|------------|---------------|------------------|--------|----------|
| 1   | -400.0 | 530.0  | 804.2                 | 313.8      | 2.520e-3      | 400.00           | 321.70 | —        |
| 2   | -200.0 | 530.0  | 804.2                 | 262.0      | 2.648e-3      | 400.00           | 321.70 | —        |
| 3   | 0.0    | 530.0  | 804.2                 | 210.1      | 2.775e-3      | 400.00           | 321.70 | —        |
| 4   | 200.0  | 530.0  | 804.2                 | 158.3      | 2.903e-3      | 400.00           | 321.70 | —        |
| 5   | 400.0  | 530.0  | 804.2                 | 106.5      | 3.030e-3      | 400.00           | 321.70 | —        |
| 6   | -400.0 | -530.0 | 804.2                 | 1337.6     | 1.971e-6      | 0.39             | 0.32   | —        |
| 7   | -200.0 | -530.0 | 804.2                 | 1285.8     | 1.295e-4      | 25.89            | 20.82  | —        |
| 8   | 0.0    | -530.0 | 804.2                 | 1233.9     | 2.570e-4      | 51.39            | 41.33  | —        |
| 9   | 200.0  | -530.0 | 804.2                 | 1182.1     | 3.845e-4      | 76.89            | 61.84  | —        |
| 10  | 400.0  | -530.0 | 804.2                 | 1130.3     | 5.120e-4      | 102.39           | 82.35  | —        |
| 11  | -680.0 | 300.0  | 804.2                 | 608.5      | 1.795e-3      | 359.09           | 288.80 | —        |
| 12  | -680.0 | 100.0  | 804.2                 | 801.7      | 1.320e-3      | 264.06           | 212.37 | —        |
| 13  | -680.0 | -100.0 | 804.2                 | 994.8      | 8.451e-4      | 169.02           | 135.94 | —        |
| 14  | -680.0 | -300.0 | 804.2                 | 1188.0     | 3.699e-4      | 73.99            | 59.50  | —        |
| 15  | 680.0  | 300.0  | 804.2                 | 256.1      | 2.662e-3      | 400.00           | 321.70 | —        |
| 16  | 680.0  | 100.0  | 804.2                 | 449.2      | 2.187e-3      | 400.00           | 321.70 | —        |
| 17  | 680.0  | -100.0 | 804.2                 | 642.4      | 1.712e-3      | 342.42           | 275.39 | —        |
| 18  | 680.0  | -300.0 | 804.2                 | 835.6      | 1.237e-3      | 247.38           | 198.96 | —        |

### Solver and admissibility evidence

|                              |              |                          |           |
|------------------------------|--------------|--------------------------|-----------|
| Strain plane $\varepsilon_0$ | 1.5162e-3    | Curvature $k_x$          | 2.3759e-6 |
| Curvature $k_y$              | 6.3748e-7    | Neutral-axis line        | 164.9806° |
| Compression depth c          | 1,338.387 mm | Iterations               | 7         |
| Relative residual            | 2.576e-12    | Max concrete compression | 0.003292  |
| Concrete limit               | 0.003300     | Max steel tension        | 0.000000  |



Steel limit **not declared**

Strain admissibility

**admissible**Governing utilization **1.5394****Vertical curve points — direct meridian at  $\beta_{eq} = 15.019^\circ$** 

One row per strain-domain station on the meridian the load point is checked against, with the Design and Nominal resultants of that station side by side. P and M $\theta$  are projected on  $\beta_{eq}$ .

| #  | Criterion                       | P design (kN) | M $\theta$ design (kN-m) | P nominal (kN) | M $\theta$ nominal (kN-m) |
|----|---------------------------------|---------------|--------------------------|----------------|---------------------------|
| 1  | Pure compression                | 17,670.3      | 0.00                     | 33,981.4       | -0.00                     |
| 2  | c/D = 3                         | 17,670.3      | 335.97                   | 33,981.4       | -0.00                     |
| 3  | c/D = 2                         | 17,670.3      | 671.95                   | 33,850.6       | 74.25                     |
| 4  | c/D = 1.5                       | 17,670.3      | 1,007.92                 | 32,775.0       | 616.42                    |
| 5  | c/D = 1.2                       | 17,670.3      | 1,343.90                 | 30,528.3       | 1,636.68                  |
| 6  | c/D = 1.1                       | 17,670.3      | 1,679.87                 | 29,098.7       | 2,258.21                  |
| 7  | c/D = 1                         | 17,558.6      | 2,034.89                 | 27,013.3       | 3,130.60                  |
| 8  | $\epsilon_s/\epsilon_y = 0$     | 16,212.6      | 2,574.90                 | 24,942.5       | 3,961.38                  |
| 9  | $\epsilon_s/\epsilon_y = 0.25$  | 13,702.1      | 3,409.27                 | 21,080.2       | 5,245.04                  |
| 10 | $\epsilon_s/\epsilon_y = 0.375$ | 12,683.1      | 3,664.28                 | 19,512.4       | 5,637.36                  |
| 11 | $\epsilon_s/\epsilon_y = 0.5$   | 11,800.1      | 3,846.28                 | 18,153.9       | 5,917.35                  |
| 12 | $\epsilon_s/\epsilon_y = 0.625$ | 11,020.8      | 3,979.53                 | 16,955.1       | 6,122.35                  |
| 13 | $\epsilon_s/\epsilon_y = 0.75$  | 10,304.6      | 4,083.62                 | 15,853.3       | 6,282.50                  |
| 14 | $\epsilon_s/\epsilon_y = 0.875$ | 9,639.9       | 4,166.53                 | 14,830.7       | 6,410.04                  |
| 15 | $\epsilon_s/\epsilon_y = 1$     | 9,016.4       | 4,234.01                 | 13,871.5       | 6,513.86                  |
| 16 | $\epsilon_s/\epsilon_y = 1.25$  | 8,348.9       | 4,504.03                 | 12,217.9       | 6,591.26                  |
| 17 | $\epsilon_s/\epsilon_y = 1.5$   | 7,814.9       | 4,678.59                 | 10,904.5       | 6,528.26                  |
| 18 | $\epsilon_s/\epsilon_y = 1.75$  | 7,366.3       | 4,792.00                 | 9,821.7        | 6,389.34                  |
| 19 | $\epsilon_s/\epsilon_y = 2$     | 6,910.3       | 4,871.99                 | 8,821.7        | 6,219.56                  |
| 20 | $\epsilon_s/\epsilon_y = 2.5$   | 6,025.0       | 4,964.47                 | 7,088.2        | 5,840.56                  |
| 21 | $\epsilon_s/\epsilon_y = 3$     | 4,802.8       | 4,618.49                 | 5,650.3        | 5,433.51                  |
| 22 | $\epsilon_s/\epsilon_y = 4$     | 2,819.9       | 3,942.47                 | 3,317.6        | 4,638.20                  |
| 23 | $\epsilon_s/\epsilon_y = 5$     | 1,437.4       | 3,365.61                 | 1,691.1        | 3,959.54                  |
| 24 | $\epsilon_s/\epsilon_y = 7.5$   | -856.6        | 2,284.53                 | -1,007.8       | 2,687.69                  |
| 25 | $\epsilon_s/\epsilon_y = 10$    | -2,301.4      | 1,539.66                 | -2,707.5       | 1,811.36                  |
| 26 | $\epsilon_s/\epsilon_y = 20$    | -4,240.2      | 438.24                   | -4,988.5       | 515.58                    |
| 27 | Pure tension                    | -4,922.0      | 0.00                     | -5,790.6       | 0.00                      |

**Fixed-P curve points — Mx-My contour at  $P_u = 24,942.9$  kN**

Cut of the fixed triangulated surface at this combination's own axial force, Design and Nominal paired on the moment direction. The last row repeats the first because the contour is drawn closed. The Design surface does not reach this axial force, so it contributes no rows and is absent from the diagram.

| #  | $\theta M$ (°) | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|----|----------------|-----------|-----------|-----------|------------|------------|------------|
| 1  | -180.000       | —         | —         | —         | -3,973.95  | -0.00      | 3,973.95   |
| 2  | -174.366       | —         | —         | —         | -3,904.69  | -385.20    | 3,923.64   |
| 3  | -165.722       | —         | —         | —         | -3,833.05  | -975.44    | 3,955.22   |
| 4  | -164.449       | —         | —         | —         | -3,810.21  | -1,060.35  | 3,955.01   |
| 5  | -158.042       | —         | —         | —         | -3,690.04  | -1,487.74  | 3,978.66   |
| 6  | -153.637       | —         | —         | —         | -3,592.34  | -1,780.36  | 4,009.31   |
| 7  | -152.796       | —         | —         | —         | -3,574.90  | -1,837.58  | 4,019.53   |
| 8  | -143.199       | —         | —         | —         | -3,300.06  | -2,468.88  | 4,121.38   |
| 9  | -141.587       | —         | —         | —         | -3,248.43  | -2,575.85  | 4,145.76   |
| 10 | -132.794       | —         | —         | —         | -2,899.05  | -3,131.30  | 4,267.26   |
| 11 | -131.336       | —         | —         | —         | -2,834.54  | -3,222.43  | 4,291.70   |
| 12 | -124.804       | —         | —         | —         | -2,469.04  | -3,551.90  | 4,325.76   |
| 13 | -122.040       | —         | —         | —         | -2,322.29  | -3,710.73  | 4,377.51   |
| 14 | -116.364       | —         | —         | —         | -1,960.31  | -3,955.30  | 4,414.44   |
| 15 | -113.389       | —         | —         | —         | -1,769.55  | -4,091.37  | 4,457.65   |
| 16 | -107.783       | —         | —         | —         | -1,379.81  | -4,302.02  | 4,517.88   |
| 17 | -105.227       | —         | —         | —         | -1,197.26  | -4,398.33  | 4,558.37   |
| 18 | -99.125        | —         | —         | —         | -736.41    | -4,584.80  | 4,643.57   |

| #  | $\theta M$ (°) | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|----|----------------|-----------|-----------|-----------|------------|------------|------------|
| 19 | -97.550        | —         | —         | —         | -613.62    | -4,629.98  | 4,670.47   |
| 20 | -90.159        | —         | —         | —         | -13.31     | -4,809.96  | 4,809.98   |
| 21 | -90.000        | —         | —         | —         | -0.01      | -4,812.16  | 4,812.16   |
| 22 | -83.802        | —         | —         | —         | 504.67     | -4,646.98  | 4,674.31   |
| 23 | -82.451        | —         | —         | —         | 613.62     | -4,629.98  | 4,670.46   |
| 24 | -77.013        | —         | —         | —         | 1,028.60   | -4,460.13  | 4,577.20   |
| 25 | -74.773        | —         | —         | —         | 1,197.26   | -4,398.32  | 4,558.36   |
| 26 | -69.371        | —         | —         | —         | 1,578.16   | -4,192.27  | 4,479.48   |
| 27 | -66.611        | —         | —         | —         | 1,769.55   | -4,091.36  | 4,457.64   |
| 28 | -60.788        | —         | —         | —         | 2,145.40   | -3,836.80  | 4,395.88   |
| 29 | -57.961        | —         | —         | —         | 2,322.27   | -3,710.72  | 4,377.48   |
| 30 | -50.273        | —         | —         | —         | 2,751.40   | -3,310.91  | 4,304.92   |
| 31 | -48.664        | —         | —         | —         | 2,834.52   | -3,222.42  | 4,291.68   |
| 32 | -40.139        | —         | —         | —         | 3,176.08   | -2,678.23  | 4,154.56   |
| 33 | -38.413        | —         | —         | —         | 3,248.44   | -2,575.86  | 4,145.78   |
| 34 | -28.209        | —         | —         | —         | 3,544.40   | -1,901.18  | 4,022.09   |
| 35 | -27.204        | —         | —         | —         | 3,574.90   | -1,837.58  | 4,019.53   |
| 36 | -21.958        | —         | —         | —         | 3,690.04   | -1,487.74  | 3,978.66   |
| 37 | -14.278        | —         | —         | —         | 3,833.05   | -975.44    | 3,955.22   |
| 38 | -6.967         | —         | —         | —         | 3,907.81   | -477.56    | 3,936.89   |
| 39 | 0.000          | —         | —         | —         | 3,973.95   | 0.00       | 3,973.95   |
| 40 | 5.634          | —         | —         | —         | 3,904.69   | 385.20     | 3,923.64   |
| 41 | 14.278         | —         | —         | —         | 3,833.05   | 975.45     | 3,955.22   |
| 42 | 15.551         | —         | —         | —         | 3,810.21   | 1,060.35   | 3,955.01   |
| 43 | 21.958         | —         | —         | —         | 3,690.04   | 1,487.76   | 3,978.67   |
| 44 | 26.363         | —         | —         | —         | 3,592.34   | 1,780.37   | 4,009.32   |
| 45 | 27.204         | —         | —         | —         | 3,574.90   | 1,837.58   | 4,019.53   |
| 46 | 36.801         | —         | —         | —         | 3,300.06   | 2,468.88   | 4,121.38   |
| 47 | 38.413         | —         | —         | —         | 3,248.43   | 2,575.85   | 4,145.76   |
| 48 | 47.206         | —         | —         | —         | 2,899.05   | 3,131.30   | 4,267.26   |
| 49 | 48.664         | —         | —         | —         | 2,834.54   | 3,222.43   | 4,291.70   |
| 50 | 55.196         | —         | —         | —         | 2,469.04   | 3,551.90   | 4,325.76   |
| 51 | 57.960         | —         | —         | —         | 2,322.29   | 3,710.73   | 4,377.51   |
| 52 | 63.636         | —         | —         | —         | 1,960.31   | 3,955.30   | 4,414.44   |
| 53 | 66.611         | —         | —         | —         | 1,769.55   | 4,091.37   | 4,457.65   |
| 54 | 72.217         | —         | —         | —         | 1,379.81   | 4,302.02   | 4,517.88   |
| 55 | 74.773         | —         | —         | —         | 1,197.26   | 4,398.33   | 4,558.37   |
| 56 | 80.875         | —         | —         | —         | 736.41     | 4,584.80   | 4,643.57   |
| 57 | 82.450         | —         | —         | —         | 613.62     | 4,629.98   | 4,670.47   |
| 58 | 89.842         | —         | —         | —         | 13.30      | 4,809.97   | 4,809.98   |
| 59 | 90.000         | —         | —         | —         | -0.00      | 4,812.17   | 4,812.17   |
| 60 | 96.198         | —         | —         | —         | -504.68    | 4,646.98   | 4,674.31   |
| 61 | 97.549         | —         | —         | —         | -613.62    | 4,629.98   | 4,670.47   |
| 62 | 102.987        | —         | —         | —         | -1,028.60  | 4,460.13   | 4,577.20   |
| 63 | 105.227        | —         | —         | —         | -1,197.26  | 4,398.32   | 4,558.36   |
| 64 | 110.629        | —         | —         | —         | -1,578.16  | 4,192.27   | 4,479.48   |
| 65 | 113.389        | —         | —         | —         | -1,769.55  | 4,091.36   | 4,457.64   |
| 66 | 119.212        | —         | —         | —         | -2,145.40  | 3,836.80   | 4,395.88   |
| 67 | 122.039        | —         | —         | —         | -2,322.27  | 3,710.72   | 4,377.48   |
| 68 | 129.727        | —         | —         | —         | -2,751.40  | 3,310.91   | 4,304.92   |
| 69 | 131.336        | —         | —         | —         | -2,834.52  | 3,222.42   | 4,291.68   |
| 70 | 139.861        | —         | —         | —         | -3,176.08  | 2,678.23   | 4,154.56   |
| 71 | 141.587        | —         | —         | —         | -3,248.44  | 2,575.86   | 4,145.78   |
| 72 | 151.791        | —         | —         | —         | -3,544.39  | 1,901.17   | 4,022.09   |
| 73 | 152.796        | —         | —         | —         | -3,574.90  | 1,837.57   | 4,019.53   |
| 74 | 158.042        | —         | —         | —         | -3,690.05  | 1,487.72   | 3,978.66   |

| #  | $\theta_M$ (°) | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|----|----------------|-----------|-----------|-----------|------------|------------|------------|
| 75 | 165.722        | —         | —         | —         | -3,833.05  | 975.44     | 3,955.22   |
| 76 | 173.033        | —         | —         | —         | -3,907.81  | 477.57     | 3,936.89   |
| 77 | -180.000       | —         | —         | —         | -3,973.95  | -0.00      | 3,973.95   |

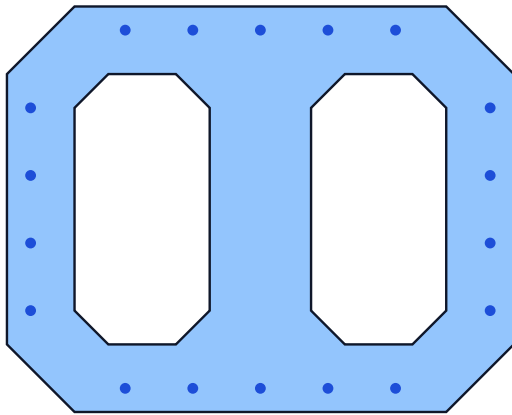
PREVIEW

## 5.2 ULS-2 same axial, half moment (inside)

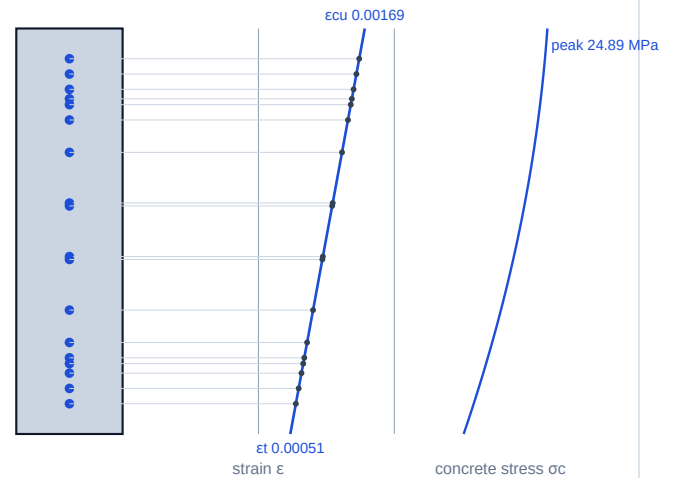
Worked section state and the points behind both curves.  $P_u = 24942.9 \text{ kN}$  ·  $M_{ux} = 1857.1 \text{ kN}\cdot\text{m}$  ·  $M_{uy} = 715.9 \text{ kN}\cdot\text{m}$  ·  $\theta_L = 21.08^\circ$

### Section state

Transverse section — neutral axis and compression zone  $\varepsilon > 0$



Longitudinal section — strain and stress over the depth  $D\theta=1456$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

### Concrete and resultants

Concrete force **22,045.7 kN**  
Concrete  $M_y$  **563.3 kN·m**  
Displaced concrete **-282.9 kN**  
Response  $P$  **24,942.9 kN**  
Response  $M_y$  **715.9 kN·m**  
Residual  $|\Delta M_x|$  **5.19e-3 N·mm**

Concrete  $M_x$  **1,476.8 kN·m**  
Steel force (gross) **3,180.1 kN**  
Steel net **2,897.2 kN**  
Response  $M_x$  **1,857.1 kN·m**  
Residual  $|\Delta P|$  **1.43e-5 N**  
Residual  $|\Delta M_y|$  **1.92e-3 N·mm**

$c = 2079.0$

### Reinforcement ledger at the solved state

| Bar | X (mm) | Y (mm) | As (mm <sup>2</sup> ) | depth (mm) | $\varepsilon$ | $\sigma_s$ (MPa) | F (kN) | in block |
|-----|--------|--------|-----------------------|------------|---------------|------------------|--------|----------|
| 1   | -400.0 | 530.0  | 804.2                 | 328.6      | 1.423e-3      | 284.64           | 228.92 | —        |
| 2   | -200.0 | 530.0  | 804.2                 | 273.6      | 1.468e-3      | 293.59           | 236.12 | —        |
| 3   | 0.0    | 530.0  | 804.2                 | 218.6      | 1.513e-3      | 302.54           | 243.32 | —        |
| 4   | 200.0  | 530.0  | 804.2                 | 163.6      | 1.557e-3      | 311.49           | 250.51 | —        |
| 5   | 400.0  | 530.0  | 804.2                 | 108.6      | 1.602e-3      | 320.43           | 257.71 | —        |
| 6   | -400.0 | -530.0 | 804.2                 | 1347.7     | 5.946e-4      | 118.92           | 95.64  | —        |
| 7   | -200.0 | -530.0 | 804.2                 | 1292.7     | 6.393e-4      | 127.87           | 102.84 | —        |
| 8   | 0.0    | -530.0 | 804.2                 | 1237.7     | 6.841e-4      | 136.81           | 110.03 | —        |
| 9   | 200.0  | -530.0 | 804.2                 | 1182.7     | 7.288e-4      | 145.76           | 117.23 | —        |
| 10  | 400.0  | -530.0 | 804.2                 | 1127.7     | 7.735e-4      | 154.71           | 124.42 | —        |
| 11  | -680.0 | 300.0  | 804.2                 | 626.8      | 1.181e-3      | 236.16           | 189.93 | —        |
| 12  | -680.0 | 100.0  | 804.2                 | 819.1      | 1.024e-3      | 204.89           | 164.78 | —        |
| 13  | -680.0 | -100.0 | 804.2                 | 1011.4     | 8.681e-4      | 173.62           | 139.63 | —        |
| 14  | -680.0 | -300.0 | 804.2                 | 1203.7     | 7.118e-4      | 142.35           | 114.49 | —        |
| 15  | 680.0  | 300.0  | 804.2                 | 252.7      | 1.485e-3      | 297.00           | 238.86 | —        |
| 16  | 680.0  | 100.0  | 804.2                 | 444.9      | 1.329e-3      | 265.73           | 213.71 | —        |
| 17  | 680.0  | -100.0 | 804.2                 | 637.2      | 1.172e-3      | 234.46           | 188.57 | —        |
| 18  | 680.0  | -300.0 | 804.2                 | 829.5      | 1.016e-3      | 203.19           | 163.42 | —        |

### Solver and admissibility evidence

Strain plane  $\varepsilon_0$  **1.0984e-3**  
Curvature  $k_y$  **2.2369e-7**  
Compression depth  $c$  **2,079.010 mm**  
Relative residual **3.635e-13**  
Concrete limit **0.003300**

Curvature  $k_x$  **7.8173e-7**  
Neutral-axis line **164.0318°**  
Iterations **5**  
Max concrete compression **0.001690**  
Max steel tension **0.000000**

Steel limit **not declared**

Strain admissibility

**admissible**Governing utilization **1.4116****Vertical curve points — direct meridian at  $\beta_{eq} = 15.968^\circ$** 

One row per strain-domain station on the meridian the load point is checked against, with the Design and Nominal resultants of that station side by side. P and M $\theta$  are projected on  $\beta_{eq}$ .

| #  | Criterion                       | P design (kN) | M $\theta$ design (kN-m) | P nominal (kN) | M $\theta$ nominal (kN-m) |
|----|---------------------------------|---------------|--------------------------|----------------|---------------------------|
| 1  | Pure compression                | 17,670.3      | 0.00                     | 33,981.4       | -0.00                     |
| 2  | c/D = 3                         | 17,670.3      | 335.03                   | 33,981.4       | -0.00                     |
| 3  | c/D = 2                         | 17,670.3      | 670.07                   | 33,854.0       | 72.79                     |
| 4  | c/D = 1.5                       | 17,670.3      | 1,005.10                 | 32,790.6       | 610.95                    |
| 5  | c/D = 1.2                       | 17,670.3      | 1,340.14                 | 30,564.8       | 1,625.29                  |
| 6  | c/D = 1.1                       | 17,670.3      | 1,675.17                 | 29,150.1       | 2,243.82                  |
| 7  | c/D = 1                         | 17,599.1      | 2,022.30                 | 27,075.5       | 3,111.23                  |
| 8  | $\epsilon_s/\epsilon_y = 0$     | 16,245.9      | 2,565.72                 | 24,993.7       | 3,947.26                  |
| 9  | $\epsilon_s/\epsilon_y = 0.25$  | 13,739.1      | 3,397.40                 | 21,137.0       | 5,226.77                  |
| 10 | $\epsilon_s/\epsilon_y = 0.375$ | 12,711.5      | 3,656.08                 | 19,556.1       | 5,624.74                  |
| 11 | $\epsilon_s/\epsilon_y = 0.5$   | 11,815.7      | 3,840.95                 | 18,178.0       | 5,909.15                  |
| 12 | $\epsilon_s/\epsilon_y = 0.625$ | 11,027.5      | 3,975.38                 | 16,965.4       | 6,115.98                  |
| 13 | $\epsilon_s/\epsilon_y = 0.75$  | 10,304.5      | 4,079.66                 | 15,853.1       | 6,276.39                  |
| 14 | $\epsilon_s/\epsilon_y = 0.875$ | 9,633.5       | 4,162.32                 | 14,820.8       | 6,403.57                  |
| 15 | $\epsilon_s/\epsilon_y = 1$     | 9,004.8       | 4,229.15                 | 13,853.6       | 6,506.39                  |
| 16 | $\epsilon_s/\epsilon_y = 1.25$  | 8,316.6       | 4,496.44                 | 12,170.7       | 6,580.15                  |
| 17 | $\epsilon_s/\epsilon_y = 1.5$   | 7,766.8       | 4,669.16                 | 10,837.4       | 6,515.11                  |
| 18 | $\epsilon_s/\epsilon_y = 1.75$  | 7,296.1       | 4,778.20                 | 9,728.1        | 6,370.94                  |
| 19 | $\epsilon_s/\epsilon_y = 2$     | 6,835.3       | 4,853.94                 | 8,725.9        | 6,196.52                  |
| 20 | $\epsilon_s/\epsilon_y = 2.5$   | 5,918.3       | 4,933.96                 | 6,962.7        | 5,804.66                  |
| 21 | $\epsilon_s/\epsilon_y = 3$     | 4,678.4       | 4,581.03                 | 5,504.1        | 5,389.45                  |
| 22 | $\epsilon_s/\epsilon_y = 4$     | 2,670.7       | 3,894.64                 | 3,142.0        | 4,581.93                  |
| 23 | $\epsilon_s/\epsilon_y = 5$     | 1,316.6       | 3,320.89                 | 1,548.9        | 3,906.93                  |
| 24 | $\epsilon_s/\epsilon_y = 7.5$   | -955.8        | 2,245.97                 | -1,124.5       | 2,642.32                  |
| 25 | $\epsilon_s/\epsilon_y = 10$    | -2,364.4      | 1,514.52                 | -2,781.6       | 1,781.78                  |
| 26 | $\epsilon_s/\epsilon_y = 20$    | -4,270.2      | 423.69                   | -5,023.8       | 498.46                    |
| 27 | Pure tension                    | -4,922.0      | 0.00                     | -5,790.6       | 0.00                      |

**Fixed-P curve points — Mx-My contour at  $P_u = 24,942.9$  kN**

Cut of the fixed triangulated surface at this combination's own axial force, Design and Nominal paired on the moment direction. The last row repeats the first because the contour is drawn closed. The Design surface does not reach this axial force, so it contributes no rows and is absent from the diagram.

| #  | $\theta M$ (°) | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|----|----------------|-----------|-----------|-----------|------------|------------|------------|
| 1  | -180.000       | —         | —         | —         | -3,973.95  | -0.00      | 3,973.95   |
| 2  | -174.366       | —         | —         | —         | -3,904.69  | -385.20    | 3,923.64   |
| 3  | -165.722       | —         | —         | —         | -3,833.05  | -975.44    | 3,955.22   |
| 4  | -164.449       | —         | —         | —         | -3,810.21  | -1,060.35  | 3,955.01   |
| 5  | -158.042       | —         | —         | —         | -3,690.04  | -1,487.74  | 3,978.66   |
| 6  | -153.637       | —         | —         | —         | -3,592.34  | -1,780.36  | 4,009.31   |
| 7  | -152.796       | —         | —         | —         | -3,574.90  | -1,837.58  | 4,019.53   |
| 8  | -143.199       | —         | —         | —         | -3,300.06  | -2,468.88  | 4,121.38   |
| 9  | -141.587       | —         | —         | —         | -3,248.43  | -2,575.85  | 4,145.76   |
| 10 | -132.794       | —         | —         | —         | -2,899.05  | -3,131.30  | 4,267.26   |
| 11 | -131.336       | —         | —         | —         | -2,834.54  | -3,222.43  | 4,291.70   |
| 12 | -124.804       | —         | —         | —         | -2,469.04  | -3,551.90  | 4,325.76   |
| 13 | -122.040       | —         | —         | —         | -2,322.29  | -3,710.73  | 4,377.51   |
| 14 | -116.364       | —         | —         | —         | -1,960.31  | -3,955.30  | 4,414.44   |
| 15 | -113.389       | —         | —         | —         | -1,769.55  | -4,091.37  | 4,457.65   |
| 16 | -107.783       | —         | —         | —         | -1,379.81  | -4,302.02  | 4,517.88   |
| 17 | -105.227       | —         | —         | —         | -1,197.26  | -4,398.33  | 4,558.37   |
| 18 | -99.125        | —         | —         | —         | -736.41    | -4,584.80  | 4,643.57   |

| #  | $\theta M$ (°) | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|----|----------------|-----------|-----------|-----------|------------|------------|------------|
| 19 | -97.550        | —         | —         | —         | -613.62    | -4,629.98  | 4,670.47   |
| 20 | -90.159        | —         | —         | —         | -13.31     | -4,809.96  | 4,809.98   |
| 21 | -90.000        | —         | —         | —         | -0.01      | -4,812.16  | 4,812.16   |
| 22 | -83.802        | —         | —         | —         | 504.67     | -4,646.98  | 4,674.31   |
| 23 | -82.451        | —         | —         | —         | 613.62     | -4,629.98  | 4,670.46   |
| 24 | -77.013        | —         | —         | —         | 1,028.60   | -4,460.13  | 4,577.20   |
| 25 | -74.773        | —         | —         | —         | 1,197.26   | -4,398.32  | 4,558.36   |
| 26 | -69.371        | —         | —         | —         | 1,578.16   | -4,192.27  | 4,479.48   |
| 27 | -66.611        | —         | —         | —         | 1,769.55   | -4,091.36  | 4,457.64   |
| 28 | -60.788        | —         | —         | —         | 2,145.40   | -3,836.80  | 4,395.88   |
| 29 | -57.961        | —         | —         | —         | 2,322.27   | -3,710.72  | 4,377.48   |
| 30 | -50.273        | —         | —         | —         | 2,751.40   | -3,310.91  | 4,304.92   |
| 31 | -48.664        | —         | —         | —         | 2,834.52   | -3,222.42  | 4,291.68   |
| 32 | -40.139        | —         | —         | —         | 3,176.08   | -2,678.23  | 4,154.56   |
| 33 | -38.413        | —         | —         | —         | 3,248.44   | -2,575.86  | 4,145.78   |
| 34 | -28.209        | —         | —         | —         | 3,544.40   | -1,901.18  | 4,022.09   |
| 35 | -27.204        | —         | —         | —         | 3,574.90   | -1,837.58  | 4,019.53   |
| 36 | -21.958        | —         | —         | —         | 3,690.04   | -1,487.74  | 3,978.66   |
| 37 | -14.278        | —         | —         | —         | 3,833.05   | -975.44    | 3,955.22   |
| 38 | -6.967         | —         | —         | —         | 3,907.81   | -477.56    | 3,936.89   |
| 39 | 0.000          | —         | —         | —         | 3,973.95   | 0.00       | 3,973.95   |
| 40 | 5.634          | —         | —         | —         | 3,904.69   | 385.20     | 3,923.64   |
| 41 | 14.278         | —         | —         | —         | 3,833.05   | 975.45     | 3,955.22   |
| 42 | 15.551         | —         | —         | —         | 3,810.21   | 1,060.35   | 3,955.01   |
| 43 | 21.958         | —         | —         | —         | 3,690.04   | 1,487.76   | 3,978.67   |
| 44 | 26.363         | —         | —         | —         | 3,592.34   | 1,780.37   | 4,009.32   |
| 45 | 27.204         | —         | —         | —         | 3,574.90   | 1,837.58   | 4,019.53   |
| 46 | 36.801         | —         | —         | —         | 3,300.06   | 2,468.88   | 4,121.38   |
| 47 | 38.413         | —         | —         | —         | 3,248.43   | 2,575.85   | 4,145.76   |
| 48 | 47.206         | —         | —         | —         | 2,899.05   | 3,131.30   | 4,267.26   |
| 49 | 48.664         | —         | —         | —         | 2,834.54   | 3,222.43   | 4,291.70   |
| 50 | 55.196         | —         | —         | —         | 2,469.04   | 3,551.90   | 4,325.76   |
| 51 | 57.960         | —         | —         | —         | 2,322.29   | 3,710.73   | 4,377.51   |
| 52 | 63.636         | —         | —         | —         | 1,960.31   | 3,955.30   | 4,414.44   |
| 53 | 66.611         | —         | —         | —         | 1,769.55   | 4,091.37   | 4,457.65   |
| 54 | 72.217         | —         | —         | —         | 1,379.81   | 4,302.02   | 4,517.88   |
| 55 | 74.773         | —         | —         | —         | 1,197.26   | 4,398.33   | 4,558.37   |
| 56 | 80.875         | —         | —         | —         | 736.41     | 4,584.80   | 4,643.57   |
| 57 | 82.450         | —         | —         | —         | 613.62     | 4,629.98   | 4,670.47   |
| 58 | 89.842         | —         | —         | —         | 13.30      | 4,809.97   | 4,809.98   |
| 59 | 90.000         | —         | —         | —         | -0.00      | 4,812.17   | 4,812.17   |
| 60 | 96.198         | —         | —         | —         | -504.68    | 4,646.98   | 4,674.31   |
| 61 | 97.549         | —         | —         | —         | -613.62    | 4,629.98   | 4,670.47   |
| 62 | 102.987        | —         | —         | —         | -1,028.60  | 4,460.13   | 4,577.20   |
| 63 | 105.227        | —         | —         | —         | -1,197.26  | 4,398.32   | 4,558.36   |
| 64 | 110.629        | —         | —         | —         | -1,578.16  | 4,192.27   | 4,479.48   |
| 65 | 113.389        | —         | —         | —         | -1,769.55  | 4,091.36   | 4,457.64   |
| 66 | 119.212        | —         | —         | —         | -2,145.40  | 3,836.80   | 4,395.88   |
| 67 | 122.039        | —         | —         | —         | -2,322.27  | 3,710.72   | 4,377.48   |
| 68 | 129.727        | —         | —         | —         | -2,751.40  | 3,310.91   | 4,304.92   |
| 69 | 131.336        | —         | —         | —         | -2,834.52  | 3,222.42   | 4,291.68   |
| 70 | 139.861        | —         | —         | —         | -3,176.08  | 2,678.23   | 4,154.56   |
| 71 | 141.587        | —         | —         | —         | -3,248.44  | 2,575.86   | 4,145.78   |
| 72 | 151.791        | —         | —         | —         | -3,544.39  | 1,901.17   | 4,022.09   |
| 73 | 152.796        | —         | —         | —         | -3,574.90  | 1,837.57   | 4,019.53   |
| 74 | 158.042        | —         | —         | —         | -3,690.05  | 1,487.72   | 3,978.66   |

| #  | $\theta_M$ (°) | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|----|----------------|-----------|-----------|-----------|------------|------------|------------|
| 75 | 165.722        | —         | —         | —         | -3,833.05  | 975.44     | 3,955.22   |
| 76 | 173.033        | —         | —         | —         | -3,907.81  | 477.57     | 3,936.89   |
| 77 | -180.000       | —         | —         | —         | -3,973.95  | -0.00      | 3,973.95   |

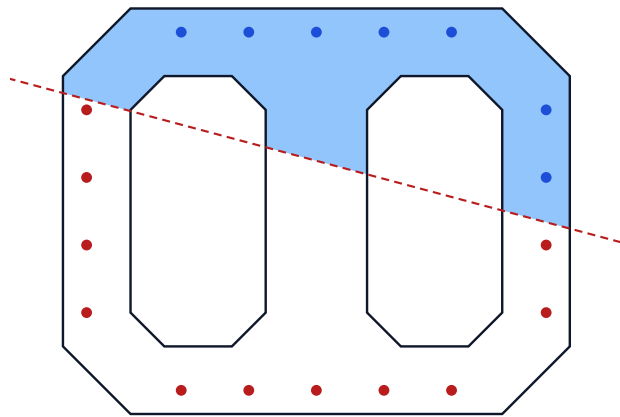
PREVIEW

### 5.3 ULS-3 low axial, high moment

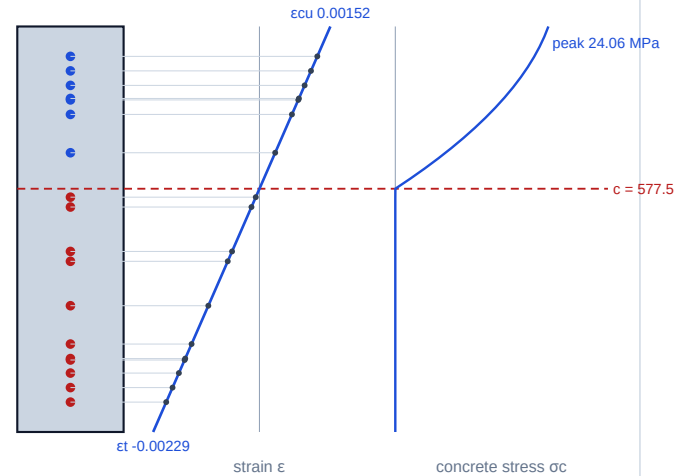
Worked section state and the points behind both curves.  $P_u = 5000.0 \text{ kN}$  ·  $M_{ux} = 4000.0 \text{ kN}\cdot\text{m}$  ·  $M_{uy} = 1500.0 \text{ kN}\cdot\text{m}$  ·  $\theta_L = 20.56^\circ$

#### Section state

Transverse section — neutral axis and compression zone  $\varepsilon > 0$



Longitudinal section — strain and stress over the depth  $D\theta=1443$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

#### Concrete and resultants

|                         |              |                         |              |
|-------------------------|--------------|-------------------------|--------------|
| Concrete force          | 6,195.6 kN   | Concrete Mx             | 2,728.2 kN·m |
| Concrete My             | 1,023.3 kN·m | Steel force (gross)     | -1,101.1 kN  |
| Displaced concrete      | -94.5 kN     | Steel net               | -1,195.6 kN  |
| Response P              | 5,000.0 kN   | Response Mx             | 4,000.0 kN·m |
| Response My             | 1,500.0 kN·m | Residual $ \Delta P $   | 5.59e-9 N    |
| Residual $ \Delta M_x $ | 0.00e+0 N·mm | Residual $ \Delta M_y $ | 9.54e-7 N·mm |

#### Reinforcement ledger at the solved state

| Bar | X (mm) | Y (mm) | As (mm <sup>2</sup> ) | depth (mm) | $\varepsilon$ | $\sigma_s$ (MPa) | F (kN)  | in block |
|-----|--------|--------|-----------------------|------------|---------------|------------------|---------|----------|
| 1   | -400.0 | 530.0  | 804.2                 | 313.0      | 6.983e-4      | 139.66           | 112.33  | —        |
| 2   | -200.0 | 530.0  | 804.2                 | 261.4      | 8.347e-4      | 166.94           | 134.26  | —        |
| 3   | 0.0    | 530.0  | 804.2                 | 209.7      | 9.711e-4      | 194.22           | 156.20  | —        |
| 4   | 200.0  | 530.0  | 804.2                 | 158.0      | 1.108e-3      | 221.50           | 178.14  | —        |
| 5   | 400.0  | 530.0  | 804.2                 | 106.4      | 1.244e-3      | 248.78           | 200.08  | —        |
| 6   | -400.0 | -530.0 | 804.2                 | 1337.1     | -2.005e-3     | -400.00          | -321.70 | —        |
| 7   | -200.0 | -530.0 | 804.2                 | 1285.4     | -1.869e-3     | -373.74          | -300.58 | —        |
| 8   | 0.0    | -530.0 | 804.2                 | 1233.7     | -1.732e-3     | -346.46          | -278.64 | —        |
| 9   | 200.0  | -530.0 | 804.2                 | 1182.1     | -1.596e-3     | -319.18          | -256.70 | —        |
| 10  | 400.0  | -530.0 | 804.2                 | 1130.4     | -1.460e-3     | -291.91          | -234.76 | —        |
| 11  | -680.0 | 300.0  | 804.2                 | 607.6      | -7.922e-5     | -15.84           | -12.74  | —        |
| 12  | -680.0 | 100.0  | 804.2                 | 800.8      | -5.893e-4     | -117.86          | -94.79  | —        |
| 13  | -680.0 | -100.0 | 804.2                 | 994.0      | -1.099e-3     | -219.88          | -176.84 | —        |
| 14  | -680.0 | -300.0 | 804.2                 | 1187.2     | -1.609e-3     | -321.89          | -258.88 | —        |
| 15  | 680.0  | 300.0  | 804.2                 | 256.2      | 8.483e-4      | 169.65           | 136.44  | —        |
| 16  | 680.0  | 100.0  | 804.2                 | 449.4      | 3.382e-4      | 67.64            | 54.40   | —        |
| 17  | 680.0  | -100.0 | 804.2                 | 642.7      | -1.719e-4     | -34.38           | -27.65  | —        |
| 18  | 680.0  | -300.0 | 804.2                 | 835.9      | -6.820e-4     | -136.40          | -109.70 | —        |

#### Solver and admissibility evidence

|                              |            |                          |           |
|------------------------------|------------|--------------------------|-----------|
| Strain plane $\varepsilon_0$ | -3.8060e-4 | Curvature $k_x$          | 2.5504e-6 |
| Curvature $k_y$              | 6.8197e-7  | Neutral-axis line        | 165.0295° |
| Compression depth c          | 577.545 mm | Iterations               | 6         |
| Relative residual            | 1.419e-16  | Max concrete compression | 0.001525  |
| Concrete limit               | 0.003300   | Max steel tension        | 0.002005  |



Steel limit **not declared**

Strain admissibility

**admissible**Governing utilization **0.8774****Vertical curve points — direct meridian at  $\beta_{eq} = 14.971^\circ$** 

One row per strain-domain station on the meridian the load point is checked against, with the Design and Nominal resultants of that station side by side. P and M $\theta$  are projected on  $\beta_{eq}$ .

| #  | Criterion                       | P design (kN) | M $\theta$ design (kN-m) | P nominal (kN) | M $\theta$ nominal (kN-m) |
|----|---------------------------------|---------------|--------------------------|----------------|---------------------------|
| 1  | Pure compression                | 17,670.3      | 0.00                     | 33,981.4       | -0.00                     |
| 2  | c/D = 3                         | 17,670.3      | 336.03                   | 33,981.4       | -0.00                     |
| 3  | c/D = 2                         | 17,670.3      | 672.05                   | 33,850.4       | 74.33                     |
| 4  | c/D = 1.5                       | 17,670.3      | 1,008.08                 | 32,774.1       | 616.71                    |
| 5  | c/D = 1.2                       | 17,670.3      | 1,344.11                 | 30,526.3       | 1,637.30                  |
| 6  | c/D = 1.1                       | 17,670.3      | 1,680.14                 | 29,095.9       | 2,259.00                  |
| 7  | c/D = 1                         | 17,556.5      | 2,035.58                 | 27,010.0       | 3,131.66                  |
| 8  | $\epsilon_s/\epsilon_y = 0$     | 16,210.8      | 2,575.41                 | 24,939.7       | 3,962.17                  |
| 9  | $\epsilon_s/\epsilon_y = 0.25$  | 13,700.2      | 3,409.90                 | 21,077.3       | 5,246.01                  |
| 10 | $\epsilon_s/\epsilon_y = 0.375$ | 12,681.6      | 3,664.72                 | 19,510.2       | 5,638.02                  |
| 11 | $\epsilon_s/\epsilon_y = 0.5$   | 11,799.3      | 3,846.56                 | 18,152.7       | 5,917.79                  |
| 12 | $\epsilon_s/\epsilon_y = 0.625$ | 11,020.5      | 3,979.75                 | 16,954.6       | 6,122.69                  |
| 13 | $\epsilon_s/\epsilon_y = 0.75$  | 10,304.7      | 4,083.84                 | 15,853.3       | 6,282.83                  |
| 14 | $\epsilon_s/\epsilon_y = 0.875$ | 9,640.3       | 4,166.76                 | 14,831.2       | 6,410.39                  |
| 15 | $\epsilon_s/\epsilon_y = 1$     | 9,017.0       | 4,234.28                 | 13,872.3       | 6,514.27                  |
| 16 | $\epsilon_s/\epsilon_y = 1.25$  | 8,350.5       | 4,504.43                 | 12,220.3       | 6,591.85                  |
| 17 | $\epsilon_s/\epsilon_y = 1.5$   | 7,817.5       | 4,679.02                 | 10,908.2       | 6,528.86                  |
| 18 | $\epsilon_s/\epsilon_y = 1.75$  | 7,369.7       | 4,792.66                 | 9,826.2        | 6,390.21                  |
| 19 | $\epsilon_s/\epsilon_y = 2$     | 6,914.2       | 4,872.95                 | 8,826.6        | 6,220.78                  |
| 20 | $\epsilon_s/\epsilon_y = 2.5$   | 6,030.5       | 4,966.09                 | 7,094.7        | 5,842.46                  |
| 21 | $\epsilon_s/\epsilon_y = 3$     | 4,809.2       | 4,620.46                 | 5,657.9        | 5,435.83                  |
| 22 | $\epsilon_s/\epsilon_y = 4$     | 2,827.8       | 3,945.03                 | 3,326.8        | 4,641.22                  |
| 23 | $\epsilon_s/\epsilon_y = 5$     | 1,443.9       | 3,368.04                 | 1,698.7        | 3,962.40                  |
| 24 | $\epsilon_s/\epsilon_y = 7.5$   | -851.2        | 2,286.67                 | -1,001.5       | 2,690.20                  |
| 25 | $\epsilon_s/\epsilon_y = 10$    | -2,298.0      | 1,541.06                 | -2,703.5       | 1,813.01                  |
| 26 | $\epsilon_s/\epsilon_y = 20$    | -4,238.6      | 439.04                   | -4,986.6       | 516.51                    |
| 27 | Pure tension                    | -4,922.0      | 0.00                     | -5,790.6       | 0.00                      |

**Fixed-P curve points — Mx-My contour at  $P_u = 5,000.0$  kN**

Cut of the fixed triangulated surface at this combination's own axial force, Design and Nominal paired on the moment direction. The last row repeats the first because the contour is drawn closed.

| #  | $\theta M$ (°) | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|----|----------------|-----------|-----------|-----------|------------|------------|------------|
| 1  | -180.000       | -4,724.11 | -0.01     | 4,724.11  | —          | —          | —          |
| 2  | -177.542       | -4,691.17 | -201.41   | 4,695.50  | —          | —          | —          |
| 3  | -166.253       | -4,548.50 | -1,112.79 | 4,682.64  | —          | —          | —          |
| 4  | -161.586       | -4,434.57 | -1,476.35 | 4,673.86  | —          | —          | —          |
| 5  | -153.000       | -4,203.75 | -2,141.93 | 4,717.99  | —          | —          | —          |
| 6  | -146.252       | -3,953.10 | -2,641.13 | 4,754.21  | —          | —          | —          |
| 7  | -141.305       | -3,687.11 | -2,953.38 | 4,724.12  | —          | —          | —          |
| 8  | -135.796       | -3,390.86 | -3,297.94 | 4,730.15  | —          | —          | —          |
| 9  | -130.955       | -3,125.05 | -3,600.66 | 4,767.67  | —          | —          | —          |
| 10 | -124.938       | -2,775.86 | -3,973.48 | 4,847.05  | —          | —          | —          |
| 11 | -121.067       | -2,556.55 | -4,243.62 | 4,954.22  | —          | —          | —          |
| 12 | -117.120       | -2,279.58 | -4,450.80 | 5,000.61  | —          | —          | —          |
| 13 | -112.138       | -1,932.24 | -4,749.58 | 5,127.57  | —          | —          | —          |
| 14 | -109.848       | -1,754.96 | -4,861.89 | 5,168.93  | —          | —          | —          |
| 15 | -103.824       | -1,288.65 | -5,236.79 | 5,393.02  | —          | —          | —          |
| 16 | -103.682       | -1,276.14 | -5,242.11 | 5,395.20  | —          | —          | —          |
| 17 | -103.399       | -1,252.45 | -5,257.60 | 5,404.72  | —          | —          | —          |
| 18 | -98.325        | -788.92   | -5,391.15 | 5,448.57  | —          | —          | —          |

| #  | $\theta M$ (°) | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|----|----------------|-----------|-----------|-----------|------------|------------|------------|
| 19 | -96.579        | -629.27   | -5,455.79 | 5,491.96  | —          | —          | —          |
| 20 | -93.606        | -346.27   | -5,494.18 | 5,505.08  | —          | —          | —          |
| 21 | -90.000        | 0.00      | -5,548.05 | 5,548.05  | —          | —          | —          |
| 22 | -88.557        | 139.23    | -5,528.41 | 5,530.16  | —          | —          | —          |
| 23 | -83.421        | 629.26    | -5,455.79 | 5,491.96  | —          | —          | —          |
| 24 | -76.601        | 1,252.41  | -5,257.62 | 5,404.73  | —          | —          | —          |
| 25 | -76.176        | 1,288.65  | -5,236.78 | 5,393.00  | —          | —          | —          |
| 26 | -75.940        | 1,308.28  | -5,223.74 | 5,385.07  | —          | —          | —          |
| 27 | -67.862        | 1,932.23  | -4,749.59 | 5,127.59  | —          | —          | —          |
| 28 | -64.567        | 2,172.23  | -4,567.83 | 5,058.03  | —          | —          | —          |
| 29 | -58.933        | 2,556.58  | -4,243.63 | 4,954.24  | —          | —          | —          |
| 30 | -53.267        | 2,893.66  | -3,877.49 | 4,838.20  | —          | —          | —          |
| 31 | -49.045        | 3,125.03  | -3,600.63 | 4,767.64  | —          | —          | —          |
| 32 | -43.480        | 3,425.33  | -3,248.20 | 4,720.56  | —          | —          | —          |
| 33 | -38.695        | 3,687.11  | -2,953.42 | 4,724.14  | —          | —          | —          |
| 34 | -36.512        | 3,792.55  | -2,807.52 | 4,718.65  | —          | —          | —          |
| 35 | -33.749        | 3,953.04  | -2,641.24 | 4,754.23  | —          | —          | —          |
| 36 | -29.563        | 4,101.37  | -2,326.44 | 4,715.24  | —          | —          | —          |
| 37 | -27.000        | 4,203.74  | -2,141.96 | 4,717.99  | —          | —          | —          |
| 38 | -22.124        | 4,319.58  | -1,756.07 | 4,662.89  | —          | —          | —          |
| 39 | -18.413        | 4,434.58  | -1,476.32 | 4,673.87  | —          | —          | —          |
| 40 | -15.389        | 4,496.67  | -1,237.70 | 4,663.90  | —          | —          | —          |
| 41 | -13.748        | 4,548.50  | -1,112.82 | 4,682.65  | —          | —          | —          |
| 42 | -7.759         | 4,586.06  | -624.85   | 4,628.43  | —          | —          | —          |
| 43 | -2.458         | 4,691.18  | -201.41   | 4,695.50  | —          | —          | —          |
| 44 | -1.292         | 4,698.69  | -105.99   | 4,699.89  | —          | —          | —          |
| 45 | 0.000          | 4,724.11  | 0.01      | 4,724.11  | —          | —          | —          |
| 46 | 2.458          | 4,691.17  | 201.41    | 4,695.50  | —          | —          | —          |
| 47 | 13.747         | 4,548.50  | 1,112.79  | 4,682.64  | —          | —          | —          |
| 48 | 18.414         | 4,434.57  | 1,476.35  | 4,673.86  | —          | —          | —          |
| 49 | 27.000         | 4,203.75  | 2,141.95  | 4,717.99  | —          | —          | —          |
| 50 | 33.747         | 3,953.10  | 2,641.12  | 4,754.21  | —          | —          | —          |
| 51 | 38.695         | 3,687.12  | 2,953.38  | 4,724.12  | —          | —          | —          |
| 52 | 44.204         | 3,390.86  | 3,297.93  | 4,730.15  | —          | —          | —          |
| 53 | 49.045         | 3,125.06  | 3,600.66  | 4,767.67  | —          | —          | —          |
| 54 | 55.062         | 2,775.86  | 3,973.48  | 4,847.05  | —          | —          | —          |
| 55 | 58.934         | 2,556.54  | 4,243.64  | 4,954.23  | —          | —          | —          |
| 56 | 62.880         | 2,279.58  | 4,450.80  | 5,000.61  | —          | —          | —          |
| 57 | 67.863         | 1,932.22  | 4,749.60  | 5,127.59  | —          | —          | —          |
| 58 | 70.152         | 1,754.96  | 4,861.90  | 5,168.94  | —          | —          | —          |
| 59 | 76.176         | 1,288.65  | 5,236.79  | 5,393.02  | —          | —          | —          |
| 60 | 76.318         | 1,276.14  | 5,242.11  | 5,395.20  | —          | —          | —          |
| 61 | 76.601         | 1,252.45  | 5,257.60  | 5,404.72  | —          | —          | —          |
| 62 | 81.675         | 788.92    | 5,391.15  | 5,448.57  | —          | —          | —          |
| 63 | 83.421         | 629.27    | 5,455.79  | 5,491.96  | —          | —          | —          |
| 64 | 86.394         | 346.27    | 5,494.18  | 5,505.08  | —          | —          | —          |
| 65 | 90.000         | -0.00     | 5,548.05  | 5,548.05  | —          | —          | —          |
| 66 | 91.443         | -139.23   | 5,528.41  | 5,530.16  | —          | —          | —          |
| 67 | 96.579         | -629.26   | 5,455.79  | 5,491.96  | —          | —          | —          |
| 68 | 103.399        | -1,252.41 | 5,257.62  | 5,404.73  | —          | —          | —          |
| 69 | 103.824        | -1,288.65 | 5,236.78  | 5,393.00  | —          | —          | —          |
| 70 | 104.060        | -1,308.28 | 5,223.74  | 5,385.07  | —          | —          | —          |
| 71 | 112.137        | -1,932.23 | 4,749.60  | 5,127.59  | —          | —          | —          |
| 72 | 115.433        | -2,172.21 | 4,567.84  | 5,058.03  | —          | —          | —          |
| 73 | 121.067        | -2,556.58 | 4,243.62  | 4,954.24  | —          | —          | —          |
| 74 | 126.733        | -2,893.67 | 3,877.47  | 4,838.19  | —          | —          | —          |

| #   | θM (°)   | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|-----|----------|-----------|-----------|-----------|------------|------------|------------|
| 75  | 130.955  | -3,125.03 | 3,600.63  | 4,767.64  | —          | —          | —          |
| 76  | 136.520  | -3,425.33 | 3,248.20  | 4,720.56  | —          | —          | —          |
| 77  | 141.305  | -3,687.11 | 2,953.42  | 4,724.14  | —          | —          | —          |
| 78  | 143.488  | -3,792.55 | 2,807.52  | 4,718.65  | —          | —          | —          |
| 79  | 146.251  | -3,953.04 | 2,641.24  | 4,754.23  | —          | —          | —          |
| 80  | 150.437  | -4,101.37 | 2,326.44  | 4,715.24  | —          | —          | —          |
| 81  | 153.000  | -4,203.74 | 2,141.95  | 4,717.99  | —          | —          | —          |
| 82  | 157.876  | -4,319.58 | 1,756.07  | 4,662.89  | —          | —          | —          |
| 83  | 161.587  | -4,434.58 | 1,476.32  | 4,673.87  | —          | —          | —          |
| 84  | 164.611  | -4,496.67 | 1,237.70  | 4,663.90  | —          | —          | —          |
| 85  | 166.252  | -4,548.50 | 1,112.82  | 4,682.65  | —          | —          | —          |
| 86  | 172.241  | -4,586.06 | 624.85    | 4,628.43  | —          | —          | —          |
| 87  | 177.542  | -4,691.18 | 201.41    | 4,695.50  | —          | —          | —          |
| 88  | 178.708  | -4,698.69 | 105.99    | 4,699.89  | —          | —          | —          |
| 89  | -180.000 | -4,724.11 | -0.01     | 4,724.11  | —          | —          | —          |
| 90  | -180.000 | —         | —         | —         | -5,265.03  | -0.04      | 5,265.03   |
| 91  | -179.840 | —         | —         | —         | -5,263.00  | -14.70     | 5,263.03   |
| 92  | -179.442 | —         | —         | —         | -5,258.15  | -51.21     | 5,258.40   |
| 93  | -170.479 | —         | —         | —         | -5,125.93  | -859.72    | 5,197.52   |
| 94  | -165.399 | —         | —         | —         | -5,054.40  | -1,316.63  | 5,223.07   |
| 95  | -152.931 | —         | —         | —         | -4,712.33  | -2,408.18  | 5,292.01   |
| 96  | -152.117 | —         | —         | —         | -4,687.41  | -2,480.05  | 5,303.06   |
| 97  | -150.022 | —         | —         | —         | -4,600.28  | -2,653.57  | 5,310.75   |
| 98  | -140.070 | —         | —         | —         | -4,166.43  | -3,487.34  | 5,433.29   |
| 99  | -130.772 | —         | —         | —         | -3,659.21  | -4,243.50  | 5,603.31   |
| 100 | -130.411 | —         | —         | —         | -3,637.99  | -4,273.03  | 5,611.94   |
| 101 | -122.746 | —         | —         | —         | -3,092.16  | -4,808.06  | 5,716.55   |
| 102 | -121.129 | —         | —         | —         | -2,979.98  | -4,934.41  | 5,764.43   |
| 103 | -115.014 | —         | —         | —         | -2,473.99  | -5,302.12  | 5,850.91   |
| 104 | -112.263 | —         | —         | —         | -2,245.39  | -5,484.80  | 5,926.62   |
| 105 | -108.161 | —         | —         | —         | -1,867.02  | -5,691.49  | 5,989.90   |
| 106 | -104.053 | —         | —         | —         | -1,479.18  | -5,909.44  | 6,091.75   |
| 107 | -102.730 | —         | —         | —         | -1,343.82  | -5,948.44  | 6,098.34   |
| 108 | -100.377 | —         | —         | —         | -1,106.84  | -6,044.39  | 6,144.90   |
| 109 | -97.546  | —         | —         | —         | -812.46    | -6,133.53  | 6,187.11   |
| 110 | -96.706  | —         | —         | —         | -725.88    | -6,173.64  | 6,216.17   |
| 111 | -92.552  | —         | —         | —         | -276.82    | -6,212.16  | 6,218.33   |
| 112 | -90.000  | —         | —         | —         | -0.00      | -6,266.37  | 6,266.37   |
| 113 | -85.640  | —         | —         | —         | 473.68     | -6,212.97  | 6,231.01   |
| 114 | -83.294  | —         | —         | —         | 725.90     | -6,173.64  | 6,216.17   |
| 115 | -79.622  | —         | —         | —         | 1,106.95   | -6,044.35  | 6,144.88   |
| 116 | -75.947  | —         | —         | —         | 1,479.17   | -5,909.45  | 6,091.76   |
| 117 | -73.114  | —         | —         | —         | 1,748.91   | -5,761.24  | 6,020.85   |
| 118 | -67.737  | —         | —         | —         | 2,245.35   | -5,484.83  | 5,926.63   |
| 119 | -61.454  | —         | —         | —         | 2,772.90   | -5,097.20  | 5,802.62   |
| 120 | -58.872  | —         | —         | —         | 2,979.95   | -4,934.42  | 5,764.43   |
| 121 | -49.953  | —         | —         | —         | 3,613.85   | -4,299.64  | 5,616.66   |
| 122 | -49.590  | —         | —         | —         | 3,638.00   | -4,273.06  | 5,611.96   |
| 123 | -42.222  | —         | —         | —         | 4,037.55   | -3,663.89  | 5,452.15   |
| 124 | -39.930  | —         | —         | —         | 4,166.39   | -3,487.39  | 5,433.30   |
| 125 | -34.706  | —         | —         | —         | 4,380.17   | -3,033.67  | 5,328.13   |
| 126 | -27.882  | —         | —         | —         | 4,687.43   | -2,479.99  | 5,303.05   |
| 127 | -27.420  | —         | —         | —         | 4,699.94   | -2,438.33  | 5,294.80   |
| 128 | -27.068  | —         | —         | —         | 4,712.35   | -2,408.13  | 5,292.01   |
| 129 | -19.051  | —         | —         | —         | 4,898.79   | -1,691.62  | 5,182.63   |
| 130 | -14.601  | —         | —         | —         | 5,054.40   | -1,316.64  | 5,223.07   |

| #   | $\theta M$ (°) | Mx design | My design | M  design | Mx nominal | My nominal | M  nominal |
|-----|----------------|-----------|-----------|-----------|------------|------------|------------|
| 131 | -11.931        | —         | —         | —         | 5,073.23   | -1,071.98  | 5,185.25   |
| 132 | -9.521         | —         | —         | —         | 5,125.93   | -859.68    | 5,197.52   |
| 133 | -5.012         | —         | —         | —         | 5,158.01   | -452.37    | 5,177.81   |
| 134 | -0.159         | —         | —         | —         | 5,263.00   | -14.64     | 5,263.02   |
| 135 | -0.061         | —         | —         | —         | 5,263.17   | -5.61      | 5,263.17   |
| 136 | 0.000          | —         | —         | —         | 5,265.03   | 0.04       | 5,265.03   |
| 137 | 0.160          | —         | —         | —         | 5,263.00   | 14.72      | 5,263.02   |
| 138 | 0.559          | —         | —         | —         | 5,258.14   | 51.31      | 5,258.39   |
| 139 | 9.521          | —         | —         | —         | 5,125.93   | 859.72     | 5,197.52   |
| 140 | 14.601         | —         | —         | —         | 5,054.40   | 1,316.62   | 5,223.07   |
| 141 | 27.069         | —         | —         | —         | 4,712.33   | 2,408.18   | 5,292.01   |
| 142 | 27.883         | —         | —         | —         | 4,687.41   | 2,480.07   | 5,303.06   |
| 143 | 29.978         | —         | —         | —         | 4,600.27   | 2,653.61   | 5,310.75   |
| 144 | 39.930         | —         | —         | —         | 4,166.43   | 3,487.34   | 5,433.29   |
| 145 | 49.228         | —         | —         | —         | 3,659.21   | 4,243.50   | 5,603.31   |
| 146 | 49.589         | —         | —         | —         | 3,637.99   | 4,273.03   | 5,611.94   |
| 147 | 57.254         | —         | —         | —         | 3,092.15   | 4,808.06   | 5,716.54   |
| 148 | 58.872         | —         | —         | —         | 2,979.96   | 4,934.42   | 5,764.43   |
| 149 | 64.986         | —         | —         | —         | 2,473.98   | 5,302.12   | 5,850.90   |
| 150 | 67.737         | —         | —         | —         | 2,245.36   | 5,484.82   | 5,926.62   |
| 151 | 71.839         | —         | —         | —         | 1,867.02   | 5,691.50   | 5,989.90   |
| 152 | 75.947         | —         | —         | —         | 1,479.18   | 5,909.44   | 6,091.75   |
| 153 | 77.270         | —         | —         | —         | 1,343.82   | 5,948.44   | 6,098.34   |
| 154 | 79.623         | —         | —         | —         | 1,106.84   | 6,044.39   | 6,144.90   |
| 155 | 82.454         | —         | —         | —         | 812.46     | 6,133.53   | 6,187.11   |
| 156 | 83.294         | —         | —         | —         | 725.88     | 6,173.64   | 6,216.17   |
| 157 | 87.448         | —         | —         | —         | 276.82     | 6,212.16   | 6,218.33   |
| 158 | 90.000         | —         | —         | —         | -0.00      | 6,266.37   | 6,266.37   |
| 159 | 94.360         | —         | —         | —         | -473.68    | 6,212.97   | 6,231.01   |
| 160 | 96.706         | —         | —         | —         | -725.90    | 6,173.64   | 6,216.17   |
| 161 | 100.378        | —         | —         | —         | -1,106.95  | 6,044.35   | 6,144.88   |
| 162 | 104.053        | —         | —         | —         | -1,479.17  | 5,909.45   | 6,091.76   |
| 163 | 106.886        | —         | —         | —         | -1,748.90  | 5,761.25   | 6,020.85   |
| 164 | 112.263        | —         | —         | —         | -2,245.34  | 5,484.83   | 5,926.63   |
| 165 | 118.547        | —         | —         | —         | -2,772.92  | 5,097.18   | 5,802.62   |
| 166 | 121.128        | —         | —         | —         | -2,979.96  | 4,934.41   | 5,764.43   |
| 167 | 130.047        | —         | —         | —         | -3,613.85  | 4,299.64   | 5,616.66   |
| 168 | 130.410        | —         | —         | —         | -3,638.00  | 4,273.06   | 5,611.96   |
| 169 | 137.778        | —         | —         | —         | -4,037.55  | 3,663.89   | 5,452.15   |
| 170 | 140.070        | —         | —         | —         | -4,166.39  | 3,487.39   | 5,433.30   |
| 171 | 145.294        | —         | —         | —         | -4,380.17  | 3,033.67   | 5,328.13   |
| 172 | 152.118        | —         | —         | —         | -4,687.43  | 2,479.98   | 5,303.05   |
| 173 | 152.580        | —         | —         | —         | -4,699.94  | 2,438.33   | 5,294.80   |
| 174 | 152.932        | —         | —         | —         | -4,712.35  | 2,408.13   | 5,292.01   |
| 175 | 160.949        | —         | —         | —         | -4,898.79  | 1,691.62   | 5,182.63   |
| 176 | 165.399        | —         | —         | —         | -5,054.40  | 1,316.63   | 5,223.07   |
| 177 | 168.069        | —         | —         | —         | -5,073.23  | 1,071.98   | 5,185.25   |
| 178 | 170.479        | —         | —         | —         | -5,125.93  | 859.68     | 5,197.52   |
| 179 | 174.988        | —         | —         | —         | -5,158.01  | 452.37     | 5,177.81   |
| 180 | 179.841        | —         | —         | —         | -5,263.01  | 14.61      | 5,263.03   |
| 181 | 179.939        | —         | —         | —         | -5,263.17  | 5.60       | 5,263.18   |
| 182 | -180.000       | —         | —         | —         | -5,265.03  | -0.04      | 5,265.03   |